An Investigation of Fractal Geometry through Affine Transformations

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Contents

1 Abstract

Fractals are complex, never-ending patterns that are self-similar, meaning they repeat the same structure at different scales. These geometric shapes are used to model irregular forms in nature, such as clouds, trees, coastlines, and blood vessels. Fractals are typically created through geometric iteration, where a shape is drawn and a specific rule is repeatedly applied to refine it, or through the use of mathematical formulas. In this paper, we explore the relationship between linear transformations, affine geometry, and fractal geometry. Using this knowledge, we attempt to create these complex geometric shapes using affine transformations embedded in Iterated Function Systems (IFS). As a result, we observe how operations from linear algebra can be used to manipulate geometric shapes and generate interesting objects such as fractals.

2 Introduction

Linear algebra is one of the most widely applied fields of mathematics. Not only can it be used to model and analyze problems arising in the physical world, but it also plays an important role in theoretical mathematics. In this paper, we investigated one such application in geometry, specifically in the geometry of fractals. Fractals are not merely structures that can be observed in the natural world, from the branching structure of our lungs to the irregular shapes of coastlines, and even in computer graphics, fractals appear in many different contexts. While we can appreciate these structures as the beauty of mother nature or the wonders of God's creation, we can also study them mathematically as interesting geometric objects.

Although the study of fractals often involves complex numbers, many of these fascinating structures can also be explored using real numbers by examining the geometric interpretations of matrix operations, particularly affine transformations. This exploration investigates how linear transformations, and affine geometry relate to fractal geometry and demonstrates the construction of multiple original fractals.

3 Self-Similarity and Basic Geometry

3.1 Sets in $\mathbb{R}^2$

Definition 3.1.1: A set in $\mathbb{R}^2$ is bounded if it can be enclosed by a suitably large circle, and closed if it contains all of its boundary points.

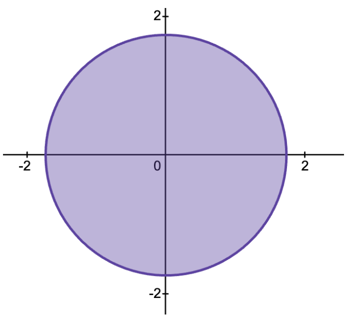


Figure 3.1.1: Closed Set

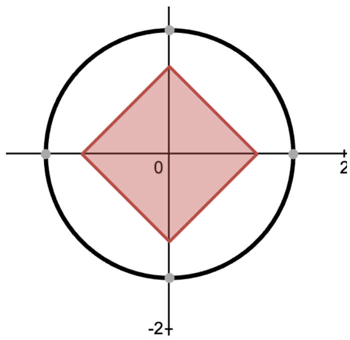


Figure 3.1.2: Bounded Set

Definition 3.1.2: Two sets are said to be *congruent* if one can be transformed into the other by a rigid motion, that is, by translations and rotations.

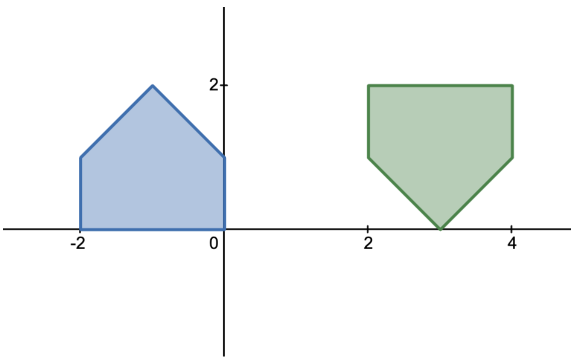


Figure 3.1.3: Congruent shapes

3.2 Self-Similar Sets

Definition 3.2.1: A closed and bounded set $S \subset \mathbb{R}^2$ is said to be *self-similar* if it can be written as

$$S = S_1 \cup S_2 \cup \cdots \cup S_k,$$

where $S_1, \ldots, S_k$ are non-overlapping sets, each congruent to a scaled copy of S with common scaling factor $0 < s < 1$.

Example 3.2.1 (Line Segment).
A line segment in $\mathbb{R}^2$ can be expressed as the union of two non-overlapping congruent line segments, each scaled by a factor of $1/2$.

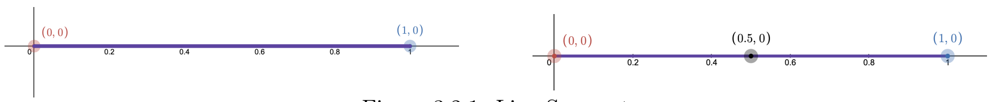


Figure 3.2.1: Line Segment

Example 3.2.2 (Unit Square).
We can express the unit square as the union of four non overlapping congruent squares, where each of the smaller squares is congruent to the original square by a scale factor of $\frac{1}{2}$.

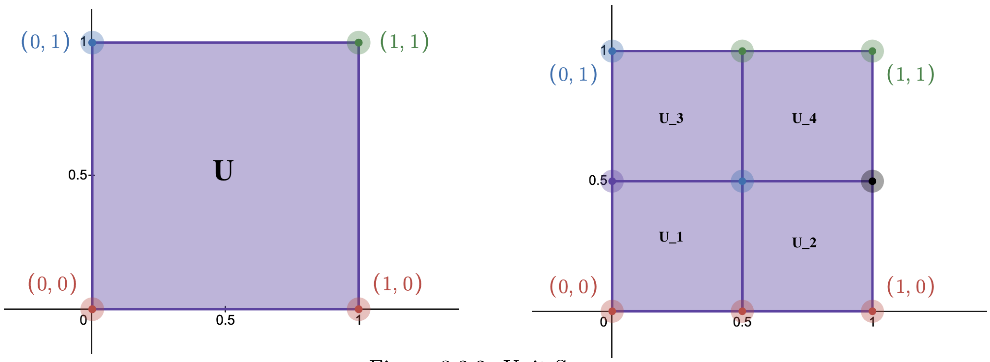


Figure 3.2.2: Unit Square

Example 3.2.3 (Sierpiński Carpet).
The Sierpinski carpet is constructed by repeatedly removing the central square from a 3×3 subdivision of a square. At each stage, the set consists of $k = 8$ scaled copies with scaling factor $s = 1/3$.

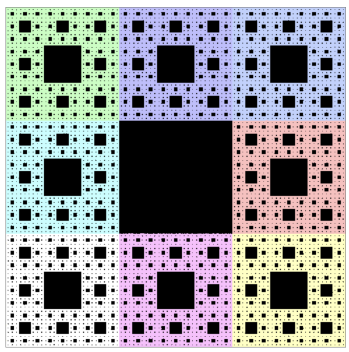


Figure 3.2.3: Sierpinski carpet

4 Dimension Theory and Fractals

4.1 Topological Dimension

Interpretation

The *topological dimension* of a set S, denoted $d_T(S)$, measures its intrinsic geometric structure, and for any set in $\mathbb{R}^n$ it must be an integer between 0 and n, inclusive. The formal definition of the topological dimension is beyond the scope of this paper. However, for better understanding we can note that a point in $\mathbb{R}^2$ has topological dimension zero, a curve in $\mathbb{R}^2$ has topological dimension one, and a region in $\mathbb{R}^2$ has topological dimension two.

Example 4.1.1: Line segment (Figure 3.2.1) : $d_T = 1$

Example 4.1.2: Unit Square (Figure 3.2.2): $d_T = 2$

Example 4.1.3: Sierpinski carpet (Figure 3.2.3): $d_T = 1$.

The reason the topological dimension of the Sierpinski carpet is 1 is that its fundamental structure resembles a web-like network of lines, due to the presence of infinitely many holes. In other words, its interior is empty because of repeatedly removing the middle square.

4.2 Hausdorff Dimension

Definition 4.2.1:
For a self-similar set with k pieces and scaling factor s,

$$d_H(S) = \frac{\ln k}{\ln(1/s)}.$$

Example 4.2.1:
Line segment (Figure 3.2.1)t: $d_H = 1$.

Example 4.2.2:
Sierpinski carpet (Figure 3.2.2): $d_H = \frac{\ln 8}{\ln 3} \approx 1.892$.

4.3 Fractals

A fractal is a complex, never-ending geometric pattern that is self-similar and a subset of Euclidean space whose Haussdorff and topological dimensions are not equal.

$$d_H(S) \neq d_T(S)$$

Example 4.3.1:
The Sierpinski carpet, demonstrated in Figure 3.2.3, is a fractal because

$$d_H(S) \approx 1.892 \quad \text{and} \quad d_T(S) = 1.$$

Now that we have established the necessary definitions and concepts for describing fractals, and have in fact defined what fractals are, we can begin to examine the applications of linear algebra in fractal geometry. In particular, we investigate how techniques developed in linear algebra can be used to generate fractals. These techniques have led to the development of algorithms that allow the process of fractal generation to be implemented on a computer.

5 Similitudes

5.1 Linear Transformations

Definition 5.1.1: Let f be a function with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$. We say that f is a transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, or that f maps from $\mathbb{R}^n$ to $\mathbb{R}^m$, which we denote by

$$f : \mathbb{R}^n \to \mathbb{R}^m.$$

In the special case where $m = n$, a transformation is sometimes called an operator on $\mathbb{R}^n$.

If $T : \mathbb{R}^2 \to \mathbb{R}^2$ is a linear operator that scales by a factor of s, and if P is a set in $\mathbb{R}^2$, then the set of images of points in P under T, $T(P)$, is called a dilation of the set P if $s > 1$ and a contraction of P if $0 < s < 1$. Hence, we can say that $T(P)$ is the set P scaled by the factor s.

5.2 Similitude

Definition 5.2.1:

A *similitude* with scale factor s is a mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ of the form

$$T\begin{pmatrix} x \\ y \end{pmatrix} = s\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix},$$

where s, θ, e, and f are real numbers. More specifically, s stretches the vector, θ rotates the vector, e and f shift the vector along the horizontal and vertical components, respectively.

Example 5.2.1 (Line Segment L).

$$T_1(x,y) = \tfrac{1}{2}(x,y), \qquad T_2(x,y) = \tfrac{1}{2}(x,y) + \left(\tfrac{1}{2}, 0\right).$$

$$L = T_1(L) \cup T_2(L).$$

Example 5.2.2 (Unit Square U). The unit square can be obtained as the union of the following four similitudes,

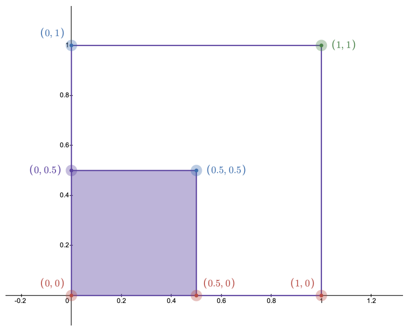
$$T_1(U) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$T_2(U) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}$$

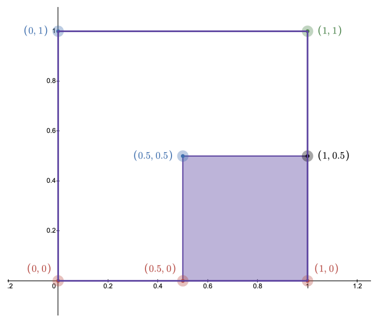
$$T_3(U) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1/2 \end{pmatrix}$$

$$T_4(U) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$$

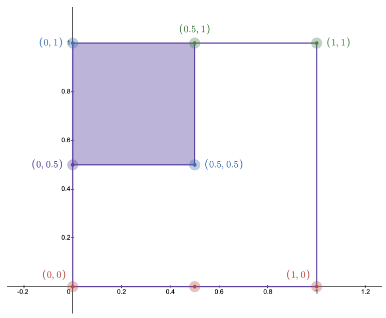
Each similitude above corresponds to one of the scaled down copies of the unit square demonstrated in Figure 5.2.1.



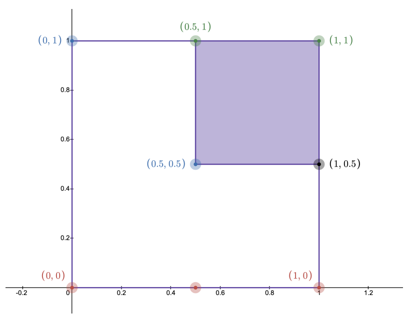
$T_1(U)$



$T_2(U)$



$T_3(U)$



$T_4(U)$

Figure 5.2.1: Similitudes of the Unit Square

Hence, the union of all the similitudes corresponds to the whole unit square demonstrated in Figure 3.2.2.

$$U = T_1(U) \cup T_2(U) \cup T_3(U) \cup T_4(U).$$

Example 5.2.3 (Sierpiński Carpet S).
For $i = 1, \dots, 8$,

$$T_i(x, y) = s \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_i \\ f_i \end{pmatrix},$$

where $\theta = 0$, and (e_i, f_i) runs through

$$(0,0),\ (\tfrac{1}{3},0),\ (\tfrac{2}{3},0),\ (0,\tfrac{1}{3}),\ (\tfrac{2}{3},\tfrac{1}{3}),\ (0,\tfrac{2}{3}),\ (\tfrac{1}{3},\tfrac{2}{3}),\ (\tfrac{2}{3},\tfrac{2}{3}).$$

$$S = \bigcup_{i=1}^{8} T_i(U).$$

6 Iterated Function Systems (IFS)

6.1 Affine Transforamtions

Definition 6.1.1: An Affine transformation is a mapping of $\mathbb{R}^2 \to \mathbb{R}^2$ of the form:

$$T \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix},$$

where a, b, c, d, e and f are scalars.

Definition 6.1.2: A contractive affine transformation is an affine map $T : \mathbb{R}^2 \to \mathbb{R}^2$,

$$T(\vec{x}) = A\vec{x} + \vec{b},$$

is *contractive* if there exists $0 < s < 1$ such that

$$\|A\vec{v}\| \leq s\|\vec{v}\| \quad \text{for all } \vec{v} \in \mathbb{R}^2.$$

Definition 6.1.3: A set of contracting affine transformations that generates a fractal by iteration is called an iterated function system (IFS).

These systems were first formalized and introduced by Australian mathematician John E. Hutchinson in his 1981 paper, *Fractals and Self-Similarity*, published in the *Indiana University Mathematics Journal* [4]. These systems create patterns by contracting a union of smaller, scaled-down copies of an initial shape. In other words, an iterated function system (IFS) maps a fractal onto itself as a collection of smaller self-similar copies, with the fractal being defined as the set of fixed points of iterated function system for affine transformations. This claim and the concept of fixed points will be discussed in greater details later.

In the affine transformations for (IFS) a, b, c, d control scaling, rotation, shear, and reflection. e and f control translation (position shift). So these transformations can be described as follows:

$$T \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} = s \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix} + \begin{pmatrix} e \\ f \end{pmatrix},$$

where s stretches the vector, θ rotates the vector, and e and f shift the vector along the horizontal and vertical components, respectively.

Now that we understand what an IFS is and how it can be used to perform mappings of one point to another, we can utilize the IFS tools to generate fractals.

The IFS tools that we utilize to generate fractals are, by convention, written in rows of numbers of the form

$$a \quad b \quad c \quad d \quad e \quad f \quad p,$$

where

$$\begin{aligned} a &= s\cos\theta \\ b &= -s\sin\theta \\ c &= s\sin\theta \\ d &= s\cos\theta \end{aligned}$$

e and f represent the components of the translation vectors for the set of affine transformations and p represents the percentage of the fractal's area generated by the corresponding transformation.

Example 6.1.1 (Sierpiński Carpet):
We have written the IFS for the Sierpinski carpet using the set of similitude discussed in Section 4 and successfully generated the Sierpinski Carpet. The corresponding IFS is given by

a	b	c	d	e	f	p
0.3333	0	0	0.3333	0.0000	0.0000	0.125
0.3333	0	0	0.3333	0.3333	0.0000	0.125
0.3333	0	0	0.3333	0.6666	0.0000	0.125
0.3333	0	0	0.3333	0.0000	0.3333	0.125
0.3333	0	0	0.3333	0.6666	0.3333	0.125
0.3333	0	0	0.3333	0.0000	0.6666	0.125
0.3333	0	0	0.3333	0.3333	0.6666	0.125
0.3333	0	0	0.3333	0.6666	0.6666	0.125

If the reader would like to verify that this IFS code generates the Sierpinski carpet, they can copy the code and paste it into the renderer available at the following link:
https://cs.lmu.edu/~ray/notes/ifs/

The reader might have noticed that in the IFS text for the Sierpinski carpet we did not include any reference to the square that we used to develop the corresponding contractive affine transformations. So why is it that the fractals generated by iterated function systems are independent of the base case? The answer can be obtained by studying the Banach's fixed point theorem.

6.2 Banach's Fixed Point Theorem

Definition 6.2.1: A fixed point of a function F is a point P such that $F(P) = P$. That is P is a fixed point of F if P is unchanged by F.

Example 6.2.1: If $f(x) = x^3$, then 0 and 1 and -1 are the fixed points of $f(x)$ because,

$$\begin{aligned} f(0) &= 0, \\ f(1) &= 1, \\ f(-1) &= -1. \end{aligned}$$

The reason behind our interest in fixed points is the fact that fractals are the fixed points of iterated function systems.

If we consider the sequence $\{P_n\}$ defined by repeated applications of F, where

$$P_{n+1} = F(P_n),$$

and suppose that

$$\lim_{n\to\infty} P_n = L,$$

then the sequence can be written as

$$
\begin{aligned}
P_1 &= F(P_0), \\
P_2 &= F^2(P_0) = F(P_1), \\
&\vdots \\
P_n &= F^n(P_0).
\end{aligned}
$$

Taking the limit as $n \to \infty$, we obtain

$$L = \lim_{n\to\infty} F^n(P_0).$$

If we evaluate F at L we get,

$$
\begin{aligned}
F(L) &= F\Big(\lim_{n\to\infty} F^n(P_0)\Big) \\
&= \lim_{n\to\infty} F(F^n(P_0)) \\
&= \lim_{n\to\infty} F^{n+1}(P_0) \\
&= \lim_{n\to\infty} F^n(P_0) \\
&= L.
\end{aligned}
$$

This implies that L is a fixed point of the function F. Additionally, the choice of P_0 irrelevant as long as the iteration converges to a fixed point. However, two important questions arise. First, is convergence guaranteed, or can the iteration diverge for certain starting points? Second, if the iteration does converge to a fixed point, is that fixed point unique, or can different starting values cause the iteration to converge to different fixed points? The following propositions are going to help us answer these questions.

Proposition 6.2.1: Let T be a continuous function and let $P_{n+1} = T(P_n)$ for all $n \geq 0$. If $L = \lim_{n\to\infty} P_n$ exists, then L is a fixed point of T.

Proof: Let T be a continuous function and let $P_{n+1} = T(P_n)$ for all $n \geq 0$. Let $L = \lim_{n\to\infty} P_n$, and consider $T(L)$.

Following the same process described above, we can conclude that $F(L) = L$. Hence, L is the fixed point of T.

♦

We have defined an iterated function system as a set of contractive affine transformations, so the type of continuous functions or mappings that we are dealing with are contractive affine transformations. If we recall the definition of contractive affine transformations, for $0 < s < 1$

$$\|A\vec{v}\| \leq s\|\vec{v}\| \quad \text{for all } \vec{v} \in \mathbb{R}^2.$$

Equivalently if T is a contractive mapping then for two points P and Q,

$$|T(P) - T(Q)| \leq s\,|P - Q| \quad \forall\, P, Q \in \mathbb{R}^2.$$

The interpretation is that contractive transformations or maps bring points closer together. As a result, Proposition 6.2.1 implies that if an iterative sequence generated by a contractive transformation converges, then it converges to a fixed point.

Proposition 6.2.2: Let T be a contractive map, the T can have at most one fixed point.

Proof: Let T be a contractive map and suppose that P and Q are both fixed points of T. Meaning, $T(P) = P$ and $T(Q) = Q$. Therefore,

$$|T(P) - T(Q)| = |P - Q|$$

Hence, T is not a contractive map, which is a contradiction to our assumption. Therefore, $P = Q$.

♦

It follows from the proposition 6.2.2, that if there exits a fixed point P for T such that $T(P) = P$ then P must be unique. This result is important in our exploration but it doesn't guarantee the existence of a fixed point. Proposition 6.2.3 is going to guarantee the existence of a fixed point for contractive maps, however before we address the proposition we need to bring in the notion of a Cauchy sequence.

Definition 6.2.2: A sequence $\{P_n\}$ is Cauchy if $\forall\ \epsilon > 0$, $\exists N \in \mathbb{N}$, such that,

$$|P_n - P_m| < \epsilon \quad \text{for all } m, n > N.$$

Key Property: All Cauchy sequences are bounded and convergent.

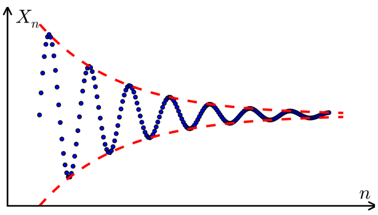


Figure 6.2.1: Graph of a Cauchy sequence

Proposition 6.2.3: Suppose that T is a contractive transformation, and that $P_{n+1} = T(P_n)$ $\forall n \geq 0$. Then $\{P_n\}$ is a Cauchy sequence for any choice of P_0.

Proof: Let T be a contractive transformation, and define $\{P_n\}$ as $P_{n+1} = T(P_n)$ $\forall n \geq 0$. Since T is a contractive transformation, there exists a constant s, $0 < s < 1$, such that,

$$\begin{aligned}
|P_{n+1} - P_n| = |T(P_n) - T(P_{n-1})| \leq s|P_n - P_{n-1}| &= s|T(P_{n-1}) - T(P_{n-2})| \\
&\vdots \\
&\leq s^n|P_1 - P_0|
\end{aligned}$$

$$\begin{aligned}
|P_{n+m+1} - P_n| &\leq |P_{n+m+1} - P_{n+m}| + \cdots + |P_{n+1} - P_n| \\
&\leq (s^{n+m} + \cdots + s^n)\,|P_1 - P_0| \\
&\leq \frac{s^n}{1-s}\,|P_1 - P_0| \\
&\leq \epsilon.
\end{aligned}$$

♦

The proof of the proposition 6.2.3 demonstrates that the points in a Cauchy sequence get arbitrarily close to one another as the sequence progresses. This brings up the notion of **complete metric space**, which is a space where every Cauchy sequence converges to a limit within that space. The standard Euclidean spaces in n dimensions are examples of complete spaces. Since fractals are defined as subsets of Euclidean space, our work remains within a complete metric space.

Now, we have everything we need to establish the existence and the uniqueness of fixed points for contractive transformation on complete space.

Banach's Fixed-Point Theorem: Suppose that T is a contractive transformation on a complete metric space, and that $P_{n+1} = T(P_n) \forall n \geq 0$. Then $L = \lim_{n\to\infty} P_n$ exists and L is the unique fixed point of T for any choice of P_0.

Proof: Since T is a contractive transformation, by proposition 6.2.3, the sequence P_n is Cauchy. Therefore, since by assumption the space is complete, $L = \lim_{n\to\infty} P_n$ exists. Hence, since T is continuous, by proposition 6.2.1, $L = \lim_{n\to\infty} P_n$ is a fixed point of T. Moreover, since T is a contractive transformation, by proposition 6.2.2, this fixed point is unique.

♦

As a result we answered both of the questioned, proposed earlier, with Banach's Fixed-Point theorem. However, another question arises that while this theorem can be applied to a single point and a single transformation, does it guarantee the same results about an iterated function system which is a collection of contractive transformations? The answer to this question lies within our investigation in the next section.

6.3 Consequences of the Banach's Fixed Point Theorem

Banach's Fixed-Point theorem can be applied in many subject areas of mathematics, such as root finding for transcendental functions, relaxation methods for solving large systems of linear equations, and algorithms for generating fractals from iterated function systems. Which for our exploration we focus on investigating its application in generating fractals from iterated function systems.

6.3.1 Compact Sets and the Hausdorff Metric

When generating fractals, we do not start with a single point, but rather with a set of points, meaning that our initial point is, in fact, a set of points. These sets are **compact sets**, and a set is compact if it is both closed and bounded. For instance, a pentagon is a compact set however a line segment is not, since it is closed but it is not bounded. Additionally, since the compact sets is a subspace of a Euclidean space, it is also complete.

When discussing convergence for Cauchy sequences we used the notion of distance, saying that as the sequence of points progresses the distance between the points becomes smaller and smaller. However, with fractals we are dealing with a sequence of sets rather than points, so we need a way to talk about the distance between sets.

Definition 6.3.1.1: The standard notion of distance for compact sets is called the Hausdorff metric and for two compact sets R and S is defined as follows,

$$H(R, S) = max\{|R - S|, |S - R|\}.$$

The Hausdorff metric on compact sets has the following properties:

$$H(R, S) \geq 0 \tag{1}$$
$$H(R, S) = 0 \Leftrightarrow R = S \tag{2}$$
$$H(R, S) = H(S, R) \tag{3}$$
$$H(R, T) \leq H(R, S) + H(S, T) \tag{4}$$

Now that we have established a notion of distance between two compact sets, we can redefine Cauchy sequences in terms of compact sets.

Definition 6.3.1.1: A sequence $\{S_n\}$ of compact sets, is Cauchy if $\forall\ \epsilon > 0$, $\exists N \in \mathbb{N}$, such that, $\forall n > N$, and $\forall m > 0$,

$$H(S_{n+m}, S_n) \leq \epsilon.$$

A Cauchy sequence of compact sets is a sequence of compact sets $\{S_n\}$ that get closer and closer as measured by the Hausdorff metric as n gets larger and larger.

Completeness Theorem:

Every Cauchy sequence of compact sets converges to a compact set.

The proof of this theorem is beyond the scope of this paper, however we can talk about the main idea behind it. If we consider a Cauchy sequence of compact sets $\{S_n\}$, we can find the set S, such that $\{S_n\}$ converges to S. Consider the Cauchy sequence of points $\{x_n\}$, then we can show that $x_n \in S_n$ as follows,

$$x \in S = \lim_{n\to\infty} S_n \Leftrightarrow x = \lim_{n\to\infty} x_n, \quad x_n \in S_n,$$

where $x = \lim_{n\to\infty} x_n$ is guaranteed to exist due to the fact that an ordinary Euclidean space is complete.

6.3.2 Fractal Theorem

Banach's fixed point theorem can be applied to contractive transformations on complete spaces, and while we have determined that the space of compact sets with the Hausdorff metric is complete, we still need to discuss the contractive transformation within this space. We already know that, the transformations used in IFS are contractive affine transformations, so we can make use of the Banach's fixed point theorem to talk about a whole set of contractive transformations not just a single one.

If we define w as a map that maps one point to another point within a Euclidean space, then we can extend w to map on sets of points S by setting,

$$w(S) = \{w(x)|x \in S\}.$$

Now let $W = \{w_1, w_2, \ldots, w_k\}$ be a collection of such maps, and extend it to map on a sets S by setting,

$$W(S) = w_1(S) \cup w_2(S) \cup \ldots \cup w_k(S).$$

W is called an iterated function system if the maps $W = \{w_1, w_2, \ldots, w_k\}$ are contractive. This does not imply that W is contractive which means that we still cannot apply Banach's fixed-point theorem.

Lemma 6.3.2.1 If $w_1, w_2, \ldots, w_k$ are each contractive transformations on Euclidean space, then $W = \{w_1, w_2, \ldots, w_k\}$ is a contractive transformation on the space of compact sets.

The intuition behind the proof of this argument is that since $w_1, w_2, \ldots, w_k$ are contractive transformations that bring pints closer together, the map $W = \{w_1, w_2, \ldots, w_k\}$ must conserve this property and bring sets closer together because sets are collections of points. A detailed proof of this lemma is beyond the scope of this paper.

The Completeness Theorem, Lemma 6.3.2.1, and the fact that the space of compact sets is a complete space, allow to finally invoke Banach's Fixed-Point theorem.

Fractal Theorem: Suppose that, $W = \{w_1, w_2, \ldots, w_k\}$ is an iterated function system, and S_0 is a compact set. Let $S_{n+1} = W(S_n)$ for all $n \geq 0$. Then $T = \lim_{n\to\infty} S_n$, exists, and T is a unique fixed point of W for any choice of S_0.

Proof: Let W be a contractive map on a complete space such that $W = \{w_1, w_2, \ldots, w_k\}$ and let $S_{n+1} = W(S_n)$ for all $n \geq 0$, with S_0 being a compact set. Then it follows from the Banach's fixed point theorem that, S_n is cauchy, therefore $T = \lim_{n\to\infty} S_n$, exists and it is a fixed of W. Additionally, T is unique.

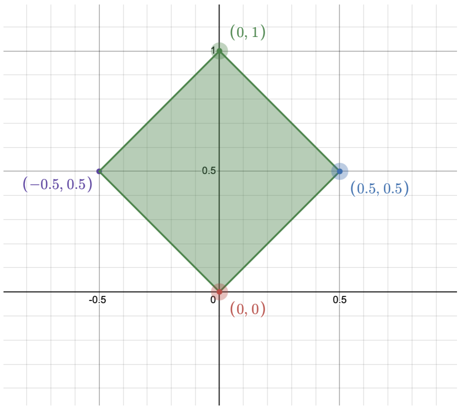
♦

At last, we have found the answer to why the fractals generated by iterated function systems are independent of the base case. This allows us to confidently begin creating our own fractals.

7 Results

7.1 Square

To create our very first original fractal we are going to consider the square R demonstrated in the figure below.



$$\begin{aligned}
AB &: (0,0) \to (0.5, 0.5),\\
BC &: (0.5, 0.5) \to (0, 1),\\
CD &: (0,1) \to (-0.5, 0.5),\\
DA &: (-0.5, 0.5) \to (0,0)
\end{aligned}$$

Figure 7.1.1: Square R.

Our plan is to create three copies of R each scaled by a factor of $s = \frac{1}{2}$. Once these copies are created, we are going to rotate and translate each copy such that the point (0,0) maps to each vertex of R excluding the point (0,0) itself, this means that each one of those vertices create the translation vectors of the transformations. The visuals of the first two iterations are demonstrated in the figure below.

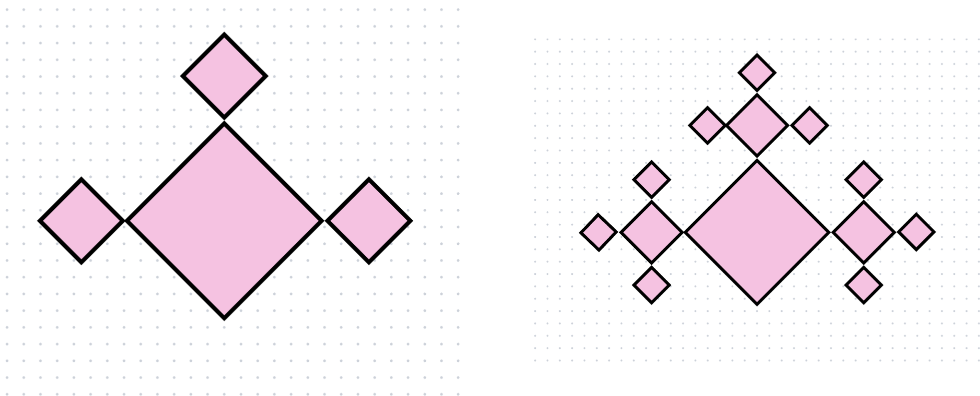


Figure 7.1.2: Fractal R^* after the first two iterations.

Since R is a square all of its angles are equal to 90°. So for right and left rotations, $\theta = -90°$ and 90° respectively. To move a copy of R up we don't need any rotations so $\theta = 0°$. Now that we have determined the angles of transformations and the corresponding translation vectors, we can form the following set of affine transformations.

$$\begin{aligned}
T_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, + \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix},\\
T_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix},\\
T_3 \begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -0.5 \\ 0.5 \end{pmatrix},
\end{aligned}$$

So the fractal R^* is determined by,

$$R^* = T_1(R) \cup T_2(R) \cup T_3(R)$$

The corresponding IFS text is as follows:

a	b	c	d	e	f	p
0.000	0.500	−0.500	0.000	0.500	0.500	0.333
0.500	0.000	0.000	0.500	0.000	1.000	0.333
0.000	−0.500	0.500	0.000	−0.500	0.500	0.333

Using this IFS text. we have successfully turned our vision into a real fractal, demonstrated in figure below.

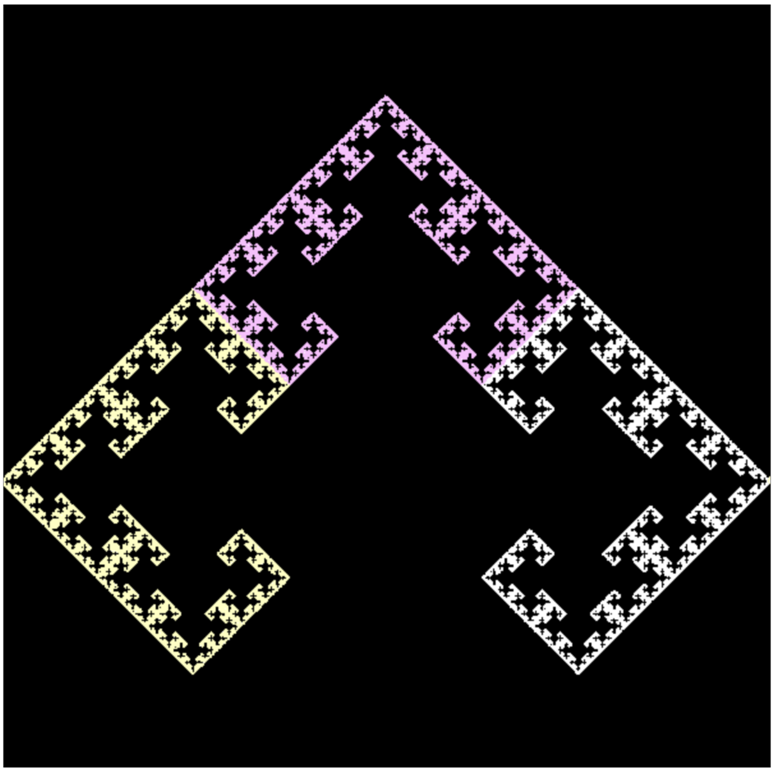


Figure 7.1.3: Original Fractal R^*.

7.2 Pentagon

Let us consider the unit pentagon P, where the length of each side is one, connecting the following points in the plane:

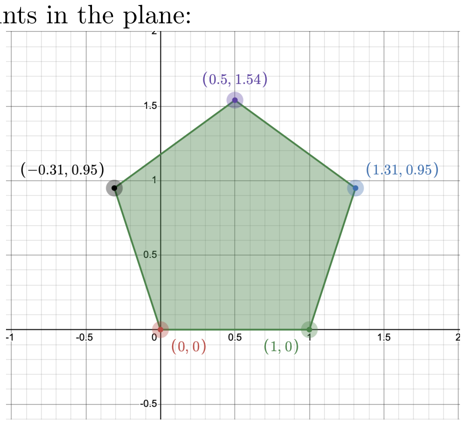


Figure 7.2.1: Pentagon P.

$$
\begin{aligned}
AB &: (0,0) \to (1,0),\\
BC &: (1,0) \to (1.31,\ 0.95),\\
CD &: (1.31,\ 0.95) \to (0.5,\ 1.54),\\
DE &: (0.5,\ 1.54) \to (-0.31,\ 0.95),\\
EA &: (-0.31,\ 0.95) \to (0,0).
\end{aligned}
$$

Our goal is to construct five copies of the unit pentagon P, each scaled by a factor of $\frac{1}{3}$. We then rotate and translate these copies so that the base of each new pentagon lies along one-third of the five sides of P. The figure below illustrates the first iteration and part of the second iteration.

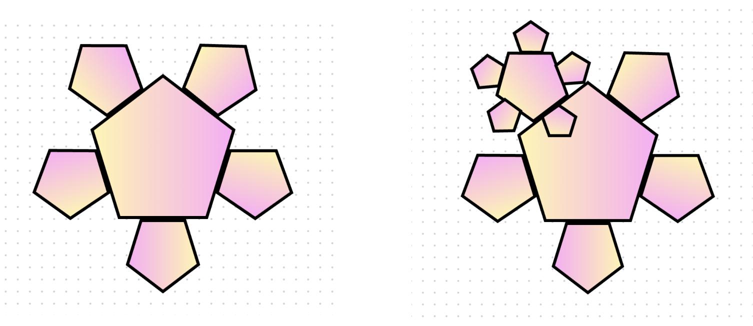


Figure 7.1.2: Early stages in the construction of P^*.

We have specified the scale factor $s = \frac{1}{3}$. However, we still need to determine the angles of rotation and the translation vectors.

The sum of the interior angles of a pentagon is 540°, so each interior angle measures 108°. To map the base AB with the side BC, we apply a rotation of $-108°$.

To map AB onto CD, we rotate by

$$108° + 108° - 180° = -36°,$$

where the subtraction of 180° accounts for flipping the pentagon so that it lies outside the original figure.

Similarly, to map AB onto DE and EA, we apply rotations of 36° and 108°, respectively. And for flipping P upside down we apply a rotation of 180°.

So our angles of rotations are as follows:

- $AB \to BC$: rotation $-108°$
- $AB \to CD$: rotation $-36°$
- $AB \to DE$: rotation 36°
- $AB \to EA$: rotation 108°
- $AB \to AB$: rotation 180°

To determine the translation vectors, we construct a parametric equation for each side of the pentagon P. Each side can be written in the form

$$p(t) = P_0 + t(P_1 - P_0), \qquad 0 \leq t \leq 1,$$

where P_0 and P_1 are the endpoints of the side.

Since the scale factor is $s = \frac{1}{3}$, the translation that places the scaled pentagon along a side of P must map the origin $(0, 0)$ to the point located at $\frac{2}{3}$ of the length of that side. Thus we evaluate each parametric equation at $t = \frac{2}{3}$.

For the side BC,

$$p_1(t) = B + t(C - B) = (1, 0) + t(0.31, 0.95).$$

Evaluating at $t = \frac{2}{3}$ gives

$$p_1\left(\tfrac{2}{3}\right) = (1, 0) + \tfrac{2}{3}(0.31, 0.95) = (1.2067,\ 0.6333).$$

For the side CD,

$$p_2(t) = C + t(D - C) = (1.31, 0.95) + t(-0.81, 0.59),$$

and

$$p_2\left(\tfrac{2}{3}\right) = (0.77,\ 1.3433).$$

For the side DE,

$$p_3(t) = D + t(E - D) = (0.5, 1.54) + t(-0.81, -0.59),$$

and

$$p_3\left(\tfrac{2}{3}\right) = (-0.04,\ 1.1467).$$

For the side EA,

$$p_4(t) = E + t(A - E) = (-0.31, 0.95) + t(0.31, -0.95),$$

and

$$p_4\left(\tfrac{2}{3}\right) = (-0.1033,\ 0.3167).$$

For the side AB,

$$p_5(t) = A + t(B - A) = (0, 0) + t(1, 0),$$

and

$$p_5\left(\tfrac{2}{3}\right) = (0.666,\, 0).$$

These points determine the translation vectors required to position the scaled pentagons along the corresponding sides of P.

Now that we have determined the scale factor s, the angles of rotation, and the translation vectors, we can construct the affine transformation matrices required for the IFS.

The corresponding affine transformations are as follows:

$$T_1\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{3}\begin{pmatrix}-0.3090 & 0.9511\\ -0.9511 & -0.3090\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}1.206\\0.633\end{pmatrix}$$
$$T_2\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{3}\begin{pmatrix}0.8090 & 0.5878\\ -0.5878 & 0.8090\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}0.77\\1.343\end{pmatrix}$$
$$T_3\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{3}\begin{pmatrix}0.8090 & -0.5878\\ 0.5878 & 0.8090\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}-0.04\\1.146\end{pmatrix}$$
$$T_4\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{3}\begin{pmatrix}-0.3090 & -0.9511\\ 0.9511 & -0.3090\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}-0.103\\0.316\end{pmatrix}$$
$$T_5\begin{pmatrix}x\\y\end{pmatrix} = \frac{1}{3}\begin{pmatrix}-1.00 & 0.00\\ 0.00 & -1.00\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}0.666\\0.000\end{pmatrix}$$

So the fractal P^* is given by

$$P^* = T_1(P) \cup T_2(P) \cup T_3(P) \cup T_4(P) \cup T_5(P).$$

The corresponding IFS text is as follows:

a	b	c	d	e	f	p
−0.1030	0.3170	−0.3170	−0.1030	1.206	0.633	0.2
0.2697	0.1959	−0.1959	0.2697	0.77	1.343	0.2
0.2697	−0.1959	0.1959	0.2697	−0.04	1.146	0.2
−0.1030	−0.3170	0.3170	−0.1030	−0.103	0.316	0.2
−0.3333	0	0	−0.3333	0.666	0	0.2

And finally we have our fractal, that we envisioned, demonstrated in figure below.

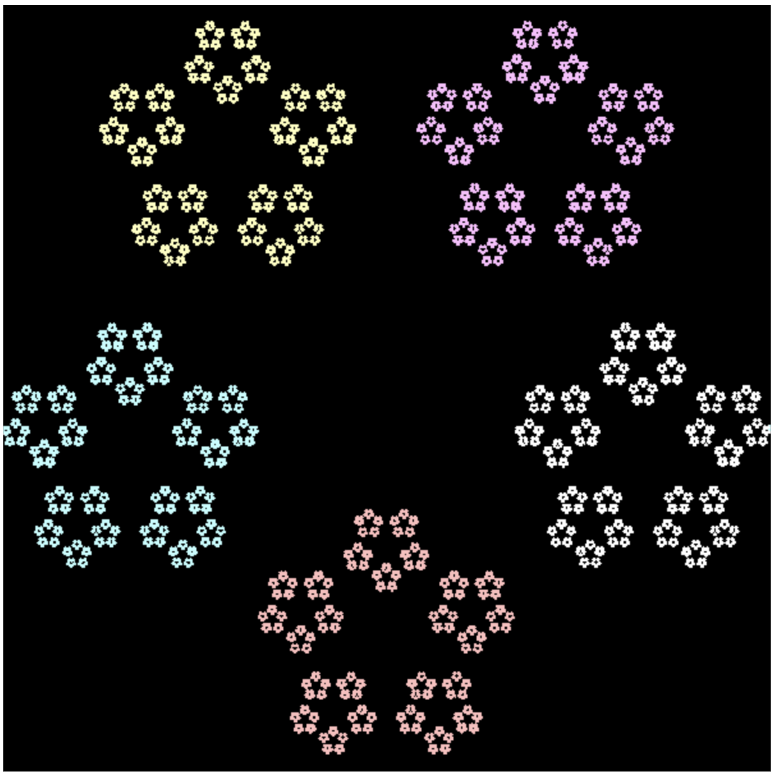


Figure 7.2.3: Original Fractal P^*.

Although P^* effectively illustrates the complexity and non-repeating patterns characteristic of a fractal, these features become even more apparent when the scale factor is increased by 0.05, so $s = 0.38$. The corresponding IFS representation is given below.

a	b	c	d	e	f	p
-0.1174	0.3614	-0.3614	-0.1174	1.206	0.633	0.2
0.3074	0.2234	-0.2234	0.3074	0.77	1.343	0.2
0.3074	-0.2234	0.2234	0.3074	-0.04	1.146	0.2
-0.1174	-0.3614	0.3614	-0.1174	-0.103	0.316	0.2
-0.3800	0	0	-0.3800	0.666	0	0.2

Accordingly, an updated version of P^* is shown in the figure below.

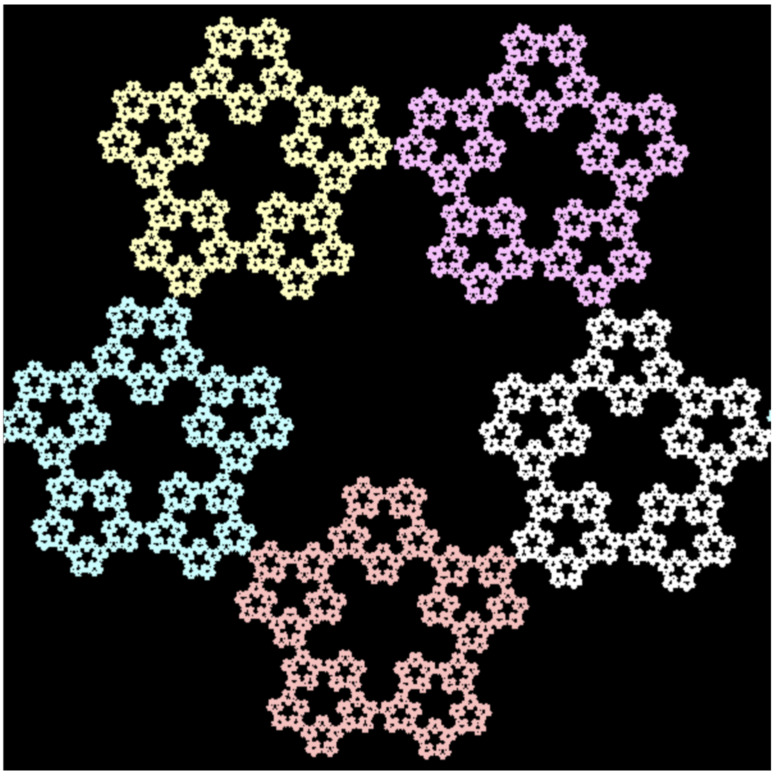


Figure 7.2.4: Original Fractal P^* with $s = 0.38$.

7.3 Regular Octagon

For our next fractal we are going to build our vision off of the regular octagon O, shown in Figure 7.3.1.

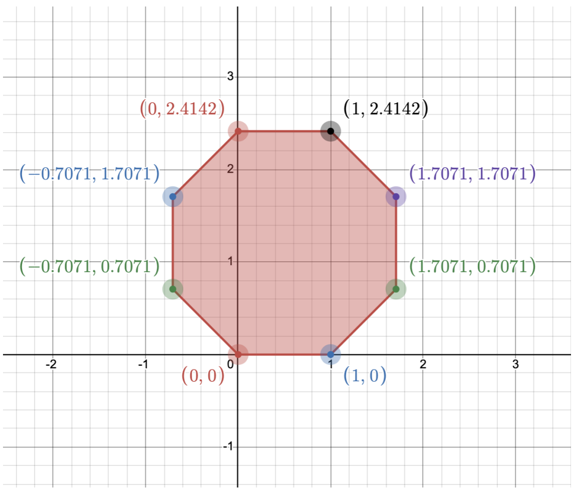


Figure 7.3.1: Octagon O.

$AB : (0, 0) \to (1, 0),$
$BC : (1, 0) \to (1.7071,\ 1.7071),$
$CD : (1.7071,\ 1.7071) \to (1.7071,\ 1.7071),$
$DE : (1.7071,\ 1.7071) \to (1,\ 2.4142),$
$EF : (1,\ 2.4142) \to (0,\ 2.4142)$
$FG : (0,\ 2.4142) \to (-0.7071,\ 1.7071),$
$GH : (-0.7071,\ 1.7071) \to (-0.7071,\ 1.7071)$
$HA : (-0.7071,\ 1.7071) \to (0, 0)$

Our approach is going to be the same as before. First we are going to create eight copies of O each scaled by a factor of $\frac{1}{3}$. Then, we are going to rotate and translate the copies so that the base of each copy lies along one-third of the eight sides of O. The figure below illustrates the first iteration.

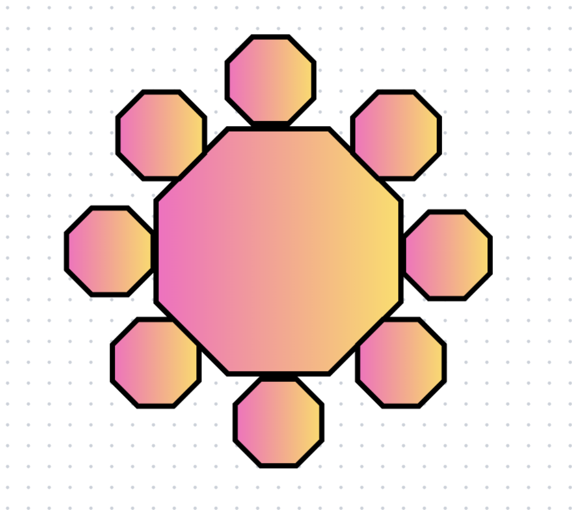


Figure 7.3.2: Fractal O^* after the first iteration.

Following the same geometric analysis as before we obtain the angles of rotation as follows:

- $AB \to BC$: rotation $-135°$
- $AB \to CD$: rotation $-90°$
- $AB \to DE$: rotation $-45°$
- $AB \to EF$: rotation $0°$
- $AB \to FG$: rotation $45°$
- $AB \to GH$: rotation $90°$
- $AB \to HA$: rotation $135°$
- $AB \to AB$: rotation $180°$

For obtaining the corresponding translation vectors, we are going to create parametric equations for each one of the eight side of O, as follows:

- $p_1(t) = B + t(C - B) = (1, 0) + t(0.7071, 0.7071)$
- $p_2(t) = C + t(D - C) = (1.7071, 0.7071) + t(0, 1)$
- $p_3(t) = D + t(E - D) = (1.7071, 1.7071) + t(-0.7071, 0.7071)$
- $p_4(t) = E + t(F - E) = (1, 2.4142) + t(-1, 0)$
- $p_5(t) = F + t(G - F) = (0, 2.4142) + t(-0.7071, -0.7071)$
- $p_6(t) = G + t(H - G) = (-0.7071, 1.7071) + t(0, -1)$
- $p_7(t) = H + t(A - H) = (-0.7071, 1.7071) + t(0.7071, -0.7071)$
- $p_8(t) = A + t(B - A) = (0, 0) + t(1, 0)$

Let $t = \frac{2}{3}$, then

- $p_1\left(\frac{2}{3}\right) = (1.4714,\ 0.4714)$

- $p_2\left(\frac{2}{3}\right) = (1.7071, 1.3737)$
- $p_3\left(\frac{2}{3}\right) = (1.2357, 2.1785)$
- $p_4\left(\frac{2}{3}\right) = (0.333, 2.4142)$
- $p_5\left(\frac{2}{3}\right) = (-0.4714, 1.9428)$
- $p_6\left(\frac{2}{3}\right) = (-0.707, 1.0404)$
- $p_7\left(\frac{2}{3}\right) = (-0.2357, 0.2357)$
- $p_8\left(\frac{2}{3}\right) = (0.666, 0)$

The corresponding affine transformations are as follows:

$$
\begin{aligned}
T_1\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}-0.7071 & 0.7071\\-0.7071 & -0.7071\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}1.4714\\0.4714\end{pmatrix}\\
T_2\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}0.000 & 1.000\\-1.000 & 0.000\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}1.7071\\1.3737\end{pmatrix}\\
T_3\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}0.7071 & 0.7071\\-0.7071 & 0.7071\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}1.2357\\2.1785\end{pmatrix}\\
T_4\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}1.000 & 0.000\\0.000 & 1.000\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}0.333\\2.4142\end{pmatrix}\\
T_5\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}0.7071 & -0.7071\\0.7071 & 0.7071\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}-0.4714\\1.9428\end{pmatrix}\\
T_6\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}0.000 & -1.000\\1.000 & 0.000\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}-0.7071\\1.0404\end{pmatrix}\\
T_7\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}-0.7071 & -0.7071\\0.7071 & -0.7071\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}-0.2357\\0.2357\end{pmatrix}\\
T_8\begin{pmatrix}x\\y\end{pmatrix} &= \frac{1}{3}\begin{pmatrix}-1.000 & 0.000\\0.000 & -1.000\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}0.666\\0.000\end{pmatrix}
\end{aligned}
$$

So the fractal O^* is given by

$$O* = T_1(O) \cup T_2(O) \cup T_3(O) \cup T_4(O) \cup T_5(O) \cup T_6(O) \cup T_7(O) \cup T_8(O).$$

The corresponding IFS text is as follows:

a	b	c	d	e	f	p
−0.2357	0.2357	−0.2357	−0.2357	1.4714	0.4714	0.1
0.0000	0.3333	−0.3333	0.0000	1.7071	1.3737	0.1
0.2357	0.2357	−0.2357	0.2357	1.2357	2.1785	0.1
0.3333	0.0000	0.0000	0.3333	0.333	2.4142	0.1
0.2357	−0.2357	0.2357	0.2357	−0.4714	1.9428	0.1
0.0000	−0.3333	0.3333	0.0000	−0.707	1.0404	0.1
−0.2357	−0.2357	0.2357	−0.2357	−0.2357	0.2357	0.1
−0.3333	0.0000	0.0000	−0.3333	0.666	0	0.1

And finally we have our fractal, that we envisioned, demonstrated in figure below.

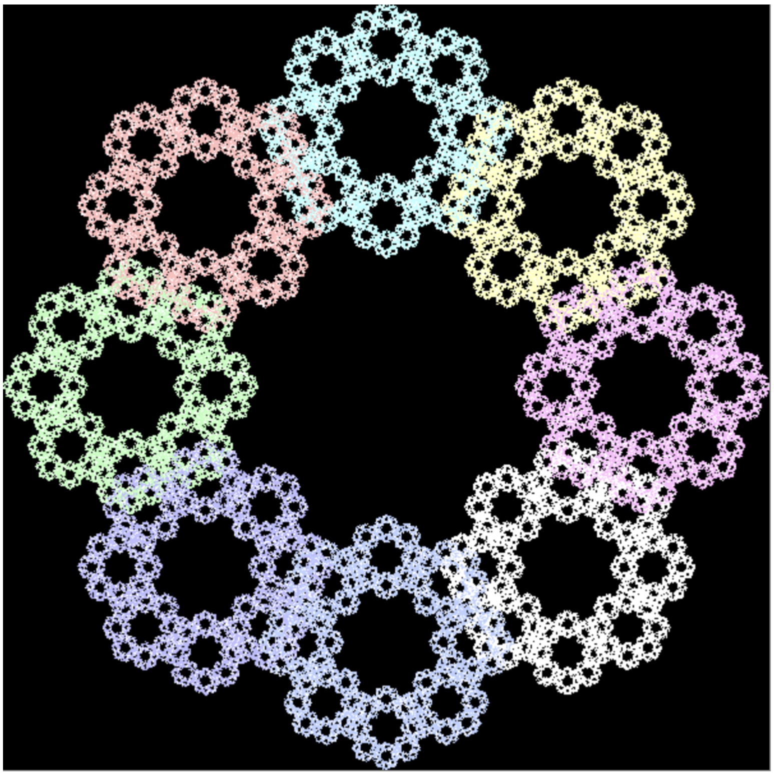


Figure 7.3.3: Original Fractal O^*.

Although O^* effectively illustrates the complexity and non-repeating patterns characteristic of a fractal, we can see a significant amount of overlap among the newly generated copies. To reduce this overlap, we set $s = 0.29$, which yields the following IFS:

a	b	c	d	e	f	p
-0.2051	0.2051	-0.2051	-0.2051	1.4714	0.4714	0.1
0.0000	0.2900	-0.2900	0.0000	1.7071	1.3737	0.1
0.2051	0.2051	-0.2051	0.2051	1.2357	2.1785	0.1
0.2900	-0.0000	0.0000	0.2900	0.333	2.4142	0.1
0.2051	-0.2051	0.2051	0.2051	-0.4714	1.9428	0.1
0.0000	-0.2900	0.2900	0.0000	-0.707	1.0404	0.1
-0.2051	-0.2051	0.2051	-0.2051	-0.2357	0.2357	0.1
-0.2900	-0.0000	0.0000	-0.2900	0.666	0	0.1

Accordingly, an updated version of O^* is shown in the Figure 7.3.3.

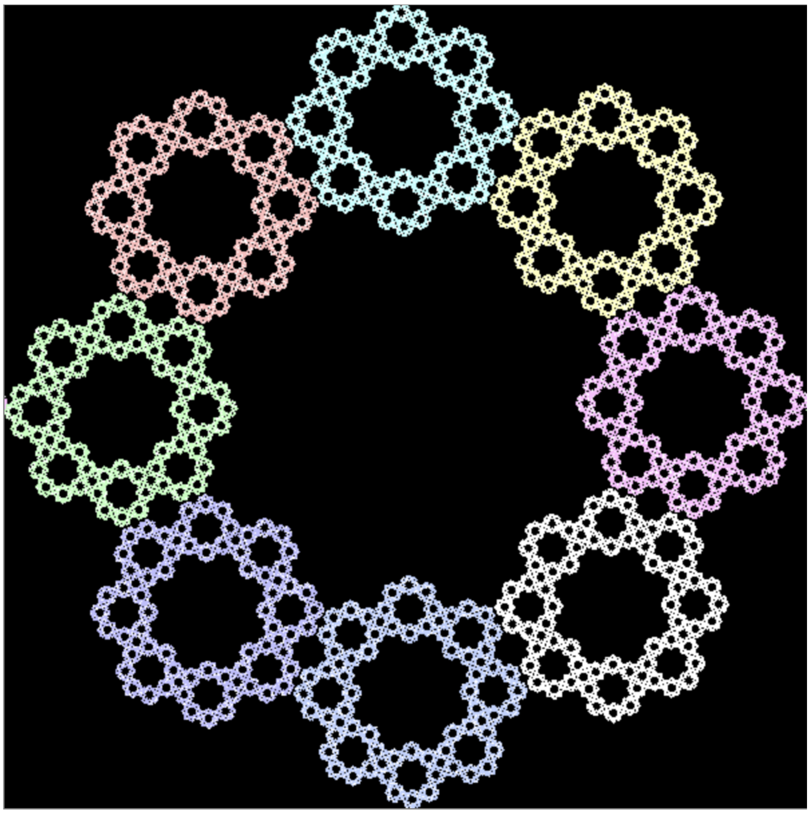


Figure 7.3.4: Original Fractal O^* with $s = 0.29$.

7.4 Irregular Octagon

To make the process more exciting, we are going to create the same fractal that we created in section 7.3 with an irregular octagon, and see how will the final result differ. Consider the Irregular Octagon M demonstrated in Figure 7.4.1.

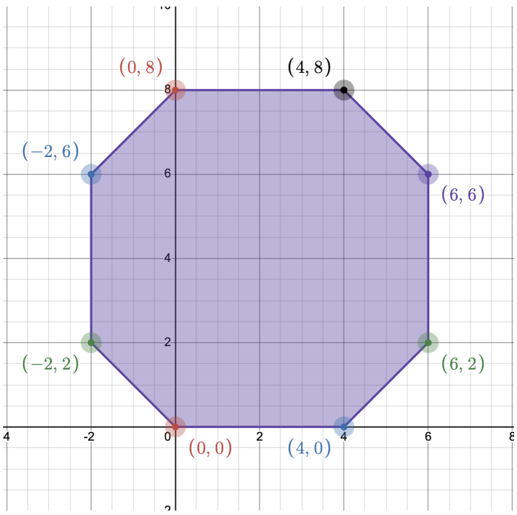


Figure 7.4.1: Irregular Octagon M.

$$
\begin{aligned}
AB &: (0,0) \to (4,0),\\
BC &: (4,0) \to (6,2),\\
CD &: (6,2) \to (6,6),\\
DE &: (6,6) \to (4,8),\\
EF &: (4,8) \to (0,8)\\
FG &: (0,8) \to (-2,6),\\
GH &: (-2,6) \to (-2,2)\\
HA &: (-2,2) \to (0,0)
\end{aligned}
$$

Angles of rotation are going to stay the same as in section 7.3, however the translation vectors will not stay the same. Additionally, since M has four sides of length two and four sides of length one, we will need to consider two different scale factors. Hence, we are going to let $s_1 = 1/4$ be the scale factor for the copies of M that will be mapped to the sides of length two, and $s_2 = \frac{\sqrt{2}}{8}$ be the scale factor for the copies of M that will be mapped to the sides of length one. The following parametric equations will be evaluate at $t = \frac{5}{12}$ to obtain the translation vectors.

- $p_1(t) = B + t(C - B) = (4, 0) + t(2, 2)$

- $p_2(t) = C + t(D - C) = (6, 2) + t(0, 4)$
- $p_3(t) = D + t(E - D) = (6, 6) + t(-2, 2)$
- $p_4(t) = E + t(F - E) = (4, 8) + t(-4, 0)$
- $p_5(t) = F + t(G - F) = (0, 8) + t(-2, -2)$
- $p_6(t) = G + t(H - G) = (-2, 6) + t(0, -4)$
- $p_7(t) = H + t(A - H) = (-2, 2) + t(2, -2)$
- $p_8(t) = A + t(B - A) = (0, 0) + t(4, 0)$

Let $t = \frac{5}{12}$, then

- $p_1\left(\frac{5}{12}\right) = (4.833, 0.833)$
- $p_2\left(\frac{5}{12}\right) = (6.00, 3.66)$
- $p_3\left(\frac{5}{12}\right) = (5.166, 6.833)$
- $p_4\left(\frac{5}{12}\right) = (2.333, 8.00)$
- $p_5\left(\frac{5}{12}\right) = (-0.833, 7.166)$
- $p_6\left(\frac{5}{12}\right) = (-2.00, 4.333)$
- $p_7\left(\frac{5}{12}\right) = (-1.166, 1.166)$
- $p_8\left(\frac{5}{12}\right) = (1.66, 0.00)$

Now we have everything we need to create the corresponding set of affine transformations.

$$
\begin{aligned}
T_1\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{\sqrt{2}}{8}\begin{pmatrix} -0.7071 & 0.7071 \\ -0.7071 & -0.7071 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4.833 \\ 0.833 \end{pmatrix} \\
T_2\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{4}\begin{pmatrix} 0.000 & 1.000 \\ -1.000 & 0.000 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 6.00 \\ 3.66 \end{pmatrix} \\
T_3\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{\sqrt{2}}{8}\begin{pmatrix} 0.7071 & 0.7071 \\ -0.7071 & 0.7071 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 5.166 \\ 6.833 \end{pmatrix} \\
T_4\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{4}\begin{pmatrix} 1.000 & 0.000 \\ 0.000 & 1.000 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 2.333 \\ 8.00 \end{pmatrix} \\
T_5\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{\sqrt{2}}{8}\begin{pmatrix} 0.7071 & -0.7071 \\ 0.7071 & 0.7071 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -0.833 \\ 7.166 \end{pmatrix} \\
T_6\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{4}\begin{pmatrix} 0.000 & -1.000 \\ 1.000 & 0.000 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -2.00 \\ 4.333 \end{pmatrix} \\
T_7\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{\sqrt{2}}{8}\begin{pmatrix} -0.7071 & -0.7071 \\ 0.7071 & -0.7071 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -1.166 \\ 1.166 \end{pmatrix} \\
T_8\begin{pmatrix} x \\ y \end{pmatrix} &= \frac{1}{4}\begin{pmatrix} -1.000 & 0.000 \\ 0.000 & -1.000 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1.666 \\ 0.000 \end{pmatrix}
\end{aligned}
$$

The corresponding IFS text is as follows:

a	b	c	d	e	f	p
−0.1250	0.1250	−0.1250	−0.1250	4.8333	0.833	0.1
0.0000	0.2500	−0.2500	0.0000	6.00	3.66	0.1
0.1250	0.1250	−0.1250	0.1250	5.1666	6.833	0.1
0.2500	−0.0000	0.0000	0.2500	2.333	8.00	0.1
0.1250	−0.1250	0.1250	0.1250	−0.833	7.166	0.1
0.0000	−0.2500	0.2500	0.0000	−2.00	4.333	0.1
−0.1250	−0.1250	0.1250	−0.1250	−1.166	1.166	0.1
−0.2500	−0.0000	0.0000	−0.2500	1.66	0.00	0.1

Hence, M^*, shown in Figure 7.4.2, can be constructed by taking the union of all the affine transformations defined above.

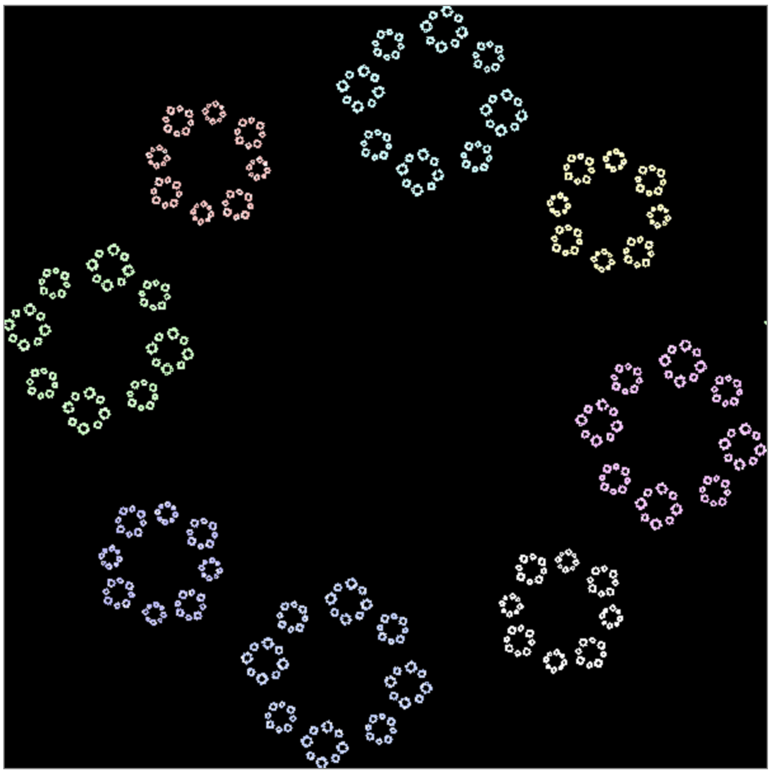


Figure 7.4.2: Original Fractal M^*.

As with all the fractals we have created before, we can play around with the scale factors, s_1 and s_2 to get a more denser fractal, however we are going to leave this exploration to the reader.

Since manually entering each of the IFS texts into the renderer would be a highly tedious process, we developed a Python program to generate the IFS text so that we can copy and paste it into the renderer. Part of the program that generates the IFS text for P^* is included in the Appendix A, however the reader may also access the whole program through the GitHub link provided in the references [5].

8 Conclusion

Linear algebra is one of the most widely applied fields of mathematics. Not only can it be used to model and analyze problems arising in the physical world, but it also plays an important role in theoretical mathematics. In this paper, we investigated one such application in geometry, specifically in the geometry of fractals. Fractals are not merely theoretical geometric objects; they are structures that can be observed in the natural world. From the branching structure of our lungs to the irregular shapes of coastlines, and even in computer graphics, fractals appear in many different contexts.

This exploration does not end here. In future work, we plan to construct additional fractals and further investigate the use of contracting transformations applied to other geometric shapes and patterns. Additionally, we are interested in investigating the construction of 3D fractals using iterated function systems. Generating 3D fractals through IFS is possible because 2D affine transformations can be extended to 3D by using (3×3) matrices and including an additional spatial dimension. This allows the same ideas used in 2D fractal generation to be applied in three dimensions.

References

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Appendix A: Python Code

```python
import math
import numpy as np

def createIFS (scalar, theta):

    transMatrix = [
        [scalar * (math.cos(theta)), -1 * scalar * math.sin(theta)],
        [scalar * (math.sin(theta)), scalar * (math.cos(theta))]
        ]
    flattened_result = " ".join(f"{item:.4f}" for sublist in transMatrix for item in sublist)
    return flattened_result

import math

angle = [-108, -36, 36, 108, -180]

transVec = [
    "1.206 0.633 0.2",
    "0.77 1.343 0.2",
    "-0.04 1.146 0.2",
    "-0.103 0.316 0.2",
    "0.666 0 0.2"
]

for i, vec in zip(angle, transVec):
    result = createIFS(1/3, math.radians(i))
    print(result, vec)
```

Figure A.1: Python code for generating the IFS text for P^*.